

Assignment 2: Version Control with Git - Initialize a new Git repository, simulate making changes to a project by adding text files, and commit those changes. Create a new branch, make further changes in this branch, and merge it back to the main branch, resolving a simulated merge conflict.

```
satya@Satyam-PC MINGW64 ~
$ cd Onedrive/Desktop

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop
$ cd Git

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git
$ git init
Initialized empty Git repository in C:/Users/satya/OneDrive/Desktop/Git/.git/

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ echo "Hello World"
Hello world

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ echo "Hello World">file1.txt

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git config --global user.name "Satyam Parida"

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git config --global user.email "satyamparida0114@gmail.com"

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git config --global user.name
Satyam Parida

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git config --global user.email
satyamparida0114@gmail.com

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file1.txt

nothing added to commit but untracked files present (use "git add" to track)

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ |
```

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git add file1.txt

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   file1.txt

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git commit -m "File Updated"
[master (root-commit) 36efecb] File Updated
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git log
commit 36efecb04042e7b6ca3689971bddaff81dc35c77 (HEAD -> master)
Author: Satyam Parida <satyamparida0114@gmail.com>
Date:   Thu Apr 3 11:02:29 2025 +0530

    File Updated

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git branch
HEAD      master

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git status
On branch master
nothing to commit, working tree clean
```

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ echo "Git" > file2.txt

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git add file2.txt
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the next time Git touches it

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git commit -m "File2 Updated"
On branch master
nothing to commit, working tree clean

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git ls-files
file1.txt
file2.txt

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git branch
HEAD      master      wipro

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git branch wipro
fatal: a branch named 'wipro' already exists

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git checkout wipro
Switched to branch 'wipro'

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ touch index.html

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git add index.html

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git status
On branch wipro
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   index.html
```

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git log
commit 33eb7f20c3a395d4e8efbdd161a1674a853cc228 (HEAD -> wipro, master)
Author: Satyam Parida <satyamparida0114@gmail.com>
Date: Thu Apr 3 11:07:42 2025 +0530
```

File2 Updated

```
commit 36efecb04042e7b6ca3689971bddaff81dc35c77
Author: Satyam Parida <satyamparida0114@gmail.com>
Date: Thu Apr 3 11:02:29 2025 +0530
```

File Updated

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git ls-files
file1.txt
file2.txt
index.html
```

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git commit -m "Index is Updated"
[wipro 19159d2] Index is Updated
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 index.html
```

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git log
commit 19159d24a5b8d522e495483c213e0b3f29568846 (HEAD -> wipro)
Author: Satyam Parida <satyamparida0114@gmail.com>
Date: Thu Apr 3 11:15:47 2025 +0530
```

Index is Updated

```
commit 33eb7f20c3a395d4e8efbdd161a1674a853cc228 (master)
Author: Satyam Parida <satyamparida0114@gmail.com>
Date: Thu Apr 3 11:07:42 2025 +0530
```

File2 Updated

```
commit 36efecb04042e7b6ca3689971bddaff81dc35c77
Author: Satyam Parida <satyamparida0114@gmail.com>
Date: Thu Apr 3 11:02:29 2025 +0530
```

File Updated

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$
```

```
satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (wipro)
$ git checkout master
Switched to branch 'master'

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git branch
HEAD      master    wipro

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git merge wipro
Updating 33eb7f2..19159d2
Fast-forward
 index.html | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 index.html

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git branch
HEAD      ORIG_HEAD  master    wipro

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ git log
commit 19159d24a5b8d522e495483c213e0b3f29568846 (HEAD -> master, wipro)
Author: Satyam Parida <satyamparida0114@gmail.com>
Date:   Thu Apr 3 11:15:47 2025 +0530

    Index is Updated

commit 33eb7f20c3a395d4e8efbdd161a1674a853cc228
Author: Satyam Parida <satyamparida0114@gmail.com>
Date:   Thu Apr 3 11:07:42 2025 +0530

    File2 Updated

commit 36efecb04042e7b6ca3689971bddaff81dc35c77
Author: Satyam Parida <satyamparida0114@gmail.com>
Date:   Thu Apr 3 11:02:29 2025 +0530

    File Updated

satya@Satyam-PC MINGW64 ~/Onedrive/Desktop/Git (master)
$ |
```